Sid Verma

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Senior Full-Stack Engineer

10+ years building products end to end: UI, APIs, data pipelines, and the infrastructure they run on. Technical ownership, from roadmap to production.

Built a social feed for **50M+ users**, infrastructure platforms at a **high-frequency trading** firm, and core parts of the open-source **Open edX** platform. Since 2024, **fractional CTO** for startups and non-profits: turning business goals into technical roadmaps, leading small teams, and running production systems. Fully remote since 2020.

Experience

Fractional CTO & Independent Consultant (2024 – Present)

Selected engagements with startups, non-profits and arts organizations, usually as the senior technical person, leading teams of up to five.

- **Tapmath**: Fractional CTO. Inherited AI-built MVPs and turned them into production-ready products: cleaned up the code, shipped new features on accelerated timelines, and set up AI agents with guardrails to automate engineering and operations work.
- **March**: Fractional CTO. Planned and delivered technical roadmaps targeting the company's biggest pain points. Stabilized the Odoo ERP behind accounting and operations, clearing tech debt and cutting the team's support load.
- **shepherd.com**: Designed and built an ingestion pipeline that collects, cleans and stores large volumes of data at low infrastructure cost.
- **India Foundation for the Arts**: Built map modules and interactive websites for exploring archival wikis.
- **ERA India**: Built low-tech, collaborative tools for farmers to record and share crop data.
- **Serendipity Arts Festival**: Led the technical build for immersive theatre and interactive installations, combining embedded hardware with real-time generative visuals.
- **HH Art Spaces**: Built the website and livestreaming infrastructure for *Open 24/7*, with AI agents that helped clients convey requirements better and guided them through feedback collection.
- Also mentored makers at Maker's Asylum and wrote science fiction about AI and society for Digital Futures Lab.

Stack: Python (Django, Flask), JavaScript (Node.js, React), Odoo, Ansible · C++ (embedded), GLSL, TouchDesigner

Senior Full-Stack Engineer › OpenCraft (2020 – 21, returned 2023 – 24)

Fully remote consultancy providing hosting, custom development and upstream contributions for Open edX, the open-source platform behind edX.

- Architected and built core Open edX components, including plugins and deployment orchestration, for clients ranging from small institutions to large global deployments.
- Upstreamed search indexing for the new content libraries backend into edx-platform (Blockstore → Elasticsearch). **PR #24530**
- Ran client infrastructure across **AWS, GCP, OpenStack and bare metal** with Terraform, Ansible and Kubernetes, and built internal tools for provisioning, deploying and monitoring it.
- Led **hiring** for the team (screening, interviews, onboarding) and mentored new developers across time zones.
- Built sprint-planning and tracking tools that made the team's delivery more predictable.

Stack: Python (Django, Flask), React, Go, Ansible, Terraform, Kubernetes, OpenStack, AWS, GCP

Full-Stack Engineer (Contract) › Shortform, Solar Accounts, McLaren College (2020 – 22)

- Built REST APIs in Python and Node.js and turned design mockups into production Vue.js apps.
- Added **end-to-end encryption** for financial data in Solar Accounts' accounting product.
- Improved API availability and response times, and cut cloud infrastructure costs.
- **McLaren College**: Designed and taught courses in programming, web development, JavaScript and Dart; built tools for running concurrent online classrooms and a backend for grading and feedback.

Stack: JavaScript (Node.js, Vue.js), Python (Django), Dart, Ansible, AWS

Full-Stack Engineer › Tower Research Capital (2017 – 19)

Internal platform team at a global high-frequency trading firm, building tools to provision, monitor and debug servers, applications and networks for trading and operations teams worldwide.

- Designed and built an internal provisioning platform (an in-house EC2) for physical and virtual servers.
- Built the firm-wide logging, monitoring and alerting platform, with fine-grained access control across teams.
- Cut response times on core platform APIs by **2–10x**.
- Redesigned the central automation framework so teams could automate both operations and approval workflows, removing redundant sign-offs and wait times.
- Led development of tools that visualize the firm's physical and logical infrastructure.

Stack: Go, Python (Django), Vue.js, React, PostgreSQL, Neo4j, Kafka, Elasticsearch, Kubernetes, Terraform

Full-Stack Engineer › Hello English (CultureAlley) (2015 – 17)

- Designed and built the in-app social feed for **50M+ users**, keeping it fast on a small infrastructure budget.
- Built a chatbot platform and a templating language that course authors used to write interactive lessons.
- Built speech recognition and pronunciation grading for Indian-accented English, running both on servers and on-device in browsers and mobile apps.
- Introduced CI/CD with automated tests, deploys and rollbacks across all apps and services.
- Turned the main web app into an offline Electron app and a PWA from a single codebase.

Stack: Python (Flask), JavaScript (Node.js), Electron, AWS, Android, Java, C++

Engineering Intern › smallcase (Tickertape) (2015 – 16)

- Sole developer of the first Stock Screener (UI, backend and data pipelines), now part of Tickertape, which has **3M+ monthly active users**.
- Wrote internal libraries for integrating vendors' REST and SOAP APIs.

Stack: Node.js, AngularJS

Skills

- **Languages:** TypeScript/JavaScript, Python, Go, SQL · also C++, Java, Rust
- **Frontend:** React, Next.js, Vue.js, D3.js, HTML/CSS
- **Backend:** Node.js, Django, Flask, REST, GraphQL, gRPC
- **Data:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Kafka, Neo4j
- **Infrastructure:** AWS, GCP, OpenStack, bare metal · Docker, Kubernetes, Terraform, Ansible, CI/CD, Linux
- **Creative tech:** TouchDesigner, GLSL, embedded C++, Blender

Education

Jaypee University of Engineering and Technology (2012 – 16)

B.Tech, Computer Science and Engineering